

Groundwater Occurrence, Hydrochemistry and Recharge Planning in Two Himalayan Catchments of Himachal Pradesh, India: A Comparative Review of the Suketi Basin and the Gaj Watershed

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Abstract

Groundwater planning in the Himalaya has to deal with steep slopes, fault-controlled aquifers, a short monsoon and growing demand. This review compares three studies from Himachal Pradesh, India. Two of them deal with the Suketi river basin in Mandi district: one describes its groundwater resource and development needs, and the other reports the chemistry of its groundwater in May and November 2009. The third maps groundwater potential in the Gaj watershed of Kangra district for 2010 and 2022 using GIS and the analytic hierarchy process, and proposes sites for recharge structures. In the Suketi basin, the alluvium of the Balh valley holds most of the usable groundwater, and only about 22% of the assessed resource is in use. The surrounding hills depend on small springs, many of which almost stop flowing before the monsoon. The water is mainly of Ca–HCO₃ type and is suitable for drinking and irrigation, except for fluoride of 4.2 mg/L in one bore well on granite and nitrate of 28–37 mg/L at two town sites. In the Gaj watershed, the area of very high potential became smaller between 2010 and 2022 as monsoon rainfall declined, and the recommended recharge sites lie on low to moderate slopes near settlements. The studies treat quantity, quality and site selection separately. We identify the main gaps in data and method and suggest how future work can link these three parts.

Keywords: Himalayan hydrogeology; Springs; Water quality; Groundwater potential; Analytic hierarchy process; Artificial recharge.

1. Introduction

Groundwater meets a large part of India's drinking and irrigation needs, and satellite data show that storage has been falling in many northern states (Rodell et al., 2009). In mountain regions, groundwater also keeps streams flowing in the dry season and feeds the springs on which many villages depend (Somers and McKenzie, 2020). In the Indian Himalaya, falling spring discharge has been reported for several decades (Valdiya and Bartarya, 1989), and a national working group has called for a systematic effort to map and revive springs (NITI Aayog, 2018).

Hill aquifers are hard to study. Slopes are steep, so much of the rain runs off before it can soak in. Thrusts and fractures break the rock into small, poorly connected aquifers. Field data are scarce, and in Himachal Pradesh official groundwater estimates are prepared only for valley areas (CGWB, 2006). Meanwhile, population, tourism and farming are growing in the foothill districts of Mandi and Kangra.

A sound management plan needs three kinds of information: how much groundwater there is and where, whether it is safe to use, and where recharge structures should be built. Three studies from the Beas catchment each provide one of these:

1. Pophare and Balpande (2015) described groundwater occurrence, wells, springs and management options in the Suketi river basin.
2. Pophare et al. (2018) studied the chemistry and quality of groundwater in the same basin.
3. Sharma et al. (2025) mapped groundwater potential in the Gaj watershed using remote sensing, GIS and multi-criteria analysis, and selected sites for artificial recharge.

This review compares these studies, brings their main results together and points out what is still missing. Each paper was read in full, and its setting, data, methods, results and recommendations were noted. Values

that we calculated from the published data are marked as derived.

2. Geological and Physiographic Setting

Both catchments drain to the Beas river and lie in the Lesser and Outer (Sub-) Himalaya of Himachal Pradesh. Their main features are listed in Table 1.

Table 1. Physical setting of the two catchments.

Feature	Suketi river basin (Pophare and Balpande, 2015; Pophare et al., 2018)	Gaj watershed (Sharma et al., 2025)
District	Mandi	Kangra
Area	422 km ² (Balh valley 79 km ² ; hills 343 km ²)	About 900 km ²
Elevation	760–2,890 m	Foothills to about 4,550 m
Rainfall	About 1,136 mm/yr, mainly July–September	About 2,600 mm/yr, mainly monsoon
Main thrusts	Mandi thrust and Main Central Thrust	Main Boundary Thrust; Jawalamukhi, Drini, Panjal and Chail thrusts
Porous aquifer	Quaternary fill of the Balh valley, up to about 370 m thick	Alluvium in the south; Siwalik sandstones and conglomerates
Hard-rock aquifer	Meta-sediments, Dhauladhar granite, Shali limestone, Subathu and Siwalik rocks	Chail formation, Dhauladhar granite

The Suketi basin has a sharp contrast in landform. In its centre lies the Balh valley, a flat intermontane basin near 790 m, filled with boulders, gravel, sand and clay. It is enclosed by hills that rise to almost 2,900 m and are made of older meta-sedimentary rocks, granite and limestone (Srikantia and Bhargava, 1998). The valley lies between the Mandi thrust and the Main Central Thrust.

The Gaj watershed is larger, higher and much wetter. It drains the southern side of the Dhauladhar range. Its upper part is made of granite and metamorphic rocks cut by thrust faults, while the lower part has Siwalik sediments and alluvium. The area is seismically active (Kumar and Mahajan, 2001), and most of its springs lie along faults and thrusts (Sharma et al., 2025).

In short, both catchments combine one porous aquifer with good storage and a much larger area of fractured hard rock with little storage.

3. Methods Used in the Three Studies

- **Field hydrogeology.** Pophare and Balpande (2015) measured the water level in 28 dug wells in May and November 2009, compiled records of 64 tube wells, and visited 93 springs, which they classed by discharge after Meinzer (1923). They also prepared maps of geology, drainage, soil, land use, lineaments and slope, and divided the basin into 53 small watersheds for planning. Drainage analysis gave a drainage density of 2.69 km/km² and an infiltration number of 10.66 (Pophare and Balpande, 2014).
- **Hydrochemistry.** Pophare et al. (2018) sampled 31 groundwater sources (dug wells, tube wells, springs and bore wells) and five river sites in both seasons of 2009. Fourteen parameters were measured by standard methods (APHA, 1998). The results were read with Piper and Gibbs diagrams (Piper, 1944; Gibbs, 1970) and with irrigation indices such as the sodium adsorption ratio (SAR), percent sodium and residual sodium carbonate (RSC) (Richards, 1954).
- **GIS and multi-criteria analysis.** Sharma et al. (2025) prepared eleven map layers from a 12.5 m ALOS PALSAR elevation model, LISS-IV satellite imagery, FAO soil data and rainfall and temperature records. Strongly related layers were removed with the variance inflation factor, weights were assigned with the analytic hierarchy process (Saaty, 1980), and the layers were combined by weighted overlay. Four maps were made (before and after the monsoon in 2010 and 2022) by changing only the rainfall layer. The maps were checked with the area under the ROC curve.

4. Groundwater Occurrence and Storage

4.1 Valley and hill aquifers

The two settings behave very differently (Table 2).

Table 2. Groundwater in the Balh valley and in the hills of the Suketi basin (Pophare and Balpande, 2015).

Parameter	Balh valley	Hills
Aquifer	Layered Quaternary gravel, sand and clay	Weathered and fractured sandstone, phyllite, granite, limestone
Main sources	Dug wells (up to 15 m); tube wells (mostly 50–125 m)	Springs; hand-pump bore wells (50–70 m)
Depth to water (2009)	Less than 12 m in May; less than 10 m in November	Not measured; many springs dry or nearly dry in May
Seasonal change	Less than 2 m in 27 of 28 wells	Spring flow rises after the monsoon, then falls quickly
Yield	Tube wells mostly 9–45 L/s; some flow at the surface	Most springs below 1 L/s (Meinzer classes VI–VIII)
Use of resource	About 22% of net availability (3.48 Mm ³ /yr)	Not assessed officially

In the Balh valley, the water table lies within 4 m of the surface in more than half of the observation wells. The net annual availability is about 3.48 Mm³, against a draft of only 0.78 Mm³, so about 2.46 Mm³ a year remains available for future use (CGWB, 2006). Some deep tube wells flow at the surface without pumping. Where the water table is within 2 m of the surface, there is a risk of waterlogging (Pophare and Balpande, 2015).

The hills are much poorer. Of the 93 springs studied, most are fracture or depression springs. In May 2009 many were dry or gave less than 0.01 L/s, and in November most gave only 0.1–0.4 L/s. A few springs on limestone or major faults, near Nalni, Nagchalla and Mandi, gave more than 10 L/s. The fine drainage texture and high infiltration number show that rain leaves the slopes quickly (Pophare and Balpande, 2014). The authors recognised three slope belts: slopes above 10° that mainly shed water, a 5–10° foot-hill belt that is the main recharge zone, and the valley floor below 5° where groundwater comes out.

In the Kangra district, records quoted by Sharma et al. (2025) show water-level falls of up to 7.16 m and a decadal change of about 6 m. Of 4,535 hand pumps in the area, 235 needed repair or replacement, partly because of low water levels at some sites.

4.2 Change in groundwater potential, 2010–2022

In the Gaj watershed, the zones of moderate to very high potential follow the main thrusts and lineaments and the southern alluvium (Sharma et al., 2025). Between 2010 and 2022, the very high class fell from 4.20% to 2.63% of the area before the monsoon, and from 4.46% to 3.20% after it (Fig. 1). Monsoon rainfall was lower in several of the later years. The maps gave AUC values of 0.82 and 0.81, which is normally taken as good accuracy.

This result needs careful reading. Rainfall was the only layer that changed between the 2010 and 2022 maps, so the maps show how changed rainfall affects the conditions for recharge. They do not measure a change in stored groundwater. The trend does, however, agree with the falling water levels in official records.

5. Groundwater Quality

5.1 Water type and controlling processes

Only the Suketi basin has chemical data (Pophare et al., 2018). The water is slightly alkaline, with pH between 6.8 and 8.3, and has a low salt content. Average conductivity is below 600 μ S/cm and average TDS below 300 mg/L for every source (Table 3). Dug wells have the highest values and springs the lowest, and all sources are more dilute after the monsoon.

Calcium and bicarbonate are the main ions, and on the Piper diagram most samples plot as Ca–HCO₃ water. Calcium and magnesium together exceed sodium and potassium, and bicarbonate exceeds chloride

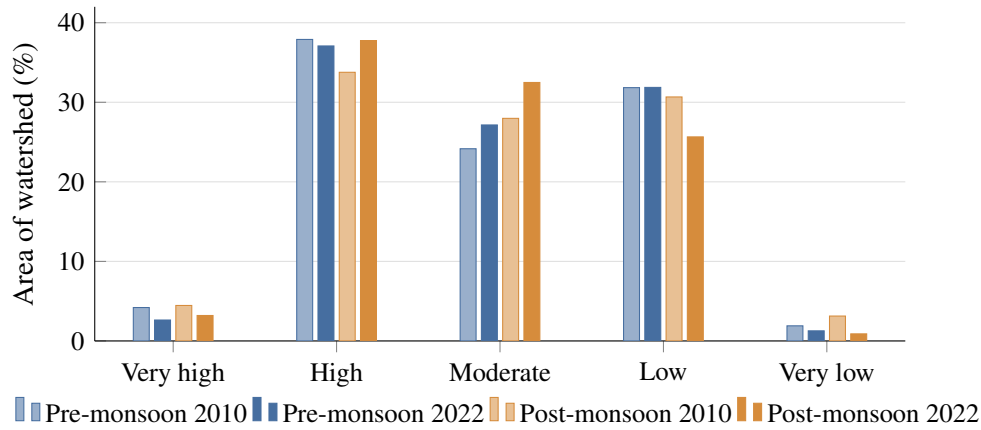


Figure 1. Share of the Gaj watershed in each groundwater potential class, drawn from the data of Sharma et al. (2025).

Table 3. Average chemistry of each water source in the Suketi basin, May / November 2009 (Pophare et al., 2018).

Source	pH	EC ($\mu\text{S/cm}$)	TDS (mg/L)	Ca (mg/L)	HCO ₃ (mg/L)	F (mg/L)
Dug wells (valley)	7.40 / 7.32	582 / 450	296 / 236	54.4 / 50.8	289 / 200	0.26 / 0.18
Tube wells (valley)	7.83 / 7.78	450 / 397	223 / 201	34.3 / 39.3	225 / 186	0.23 / 0.18
Springs (hills)	7.83 / 7.71	350 / 296	181 / 153	36.3 / 34.2	177 / 148	0.17 / 0.10
Bore wells (hills)	7.91 / 7.70	506 / 399	253 / 201	40.0 / 38.9	254 / 185	0.68 / 0.36
River water	8.14 / 7.96	384 / 344	195 / 178	34.8 / 35.2	184 / 157	0.28 / 0.17

The bore-well fluoride average is raised by one well at Pabu (4.2 / 2.2 mg/L).

and sulphate. On the Gibbs diagram most samples fall in the rainfall-dominated field. This means that water moves through the steep ground quickly and has little time to dissolve minerals from the rock.

Springs and rivers have almost the same ionic make-up in both seasons. This indicates that shallow aquifers in the hills supply the base flow of the Suketi river and its tributaries.

5.2 Local problems and irrigation use

Two local problems were found.

1. **Fluoride.** Bore well B-6 at Pabu, on granite, had 4.2 mg/L of fluoride in May and 2.2 mg/L in November. Both values are above the limit of 1.5 mg/L (BIS, 2012; WHO, 2017). The authors link this to fluoride released from the granite during long contact between water and rock.
2. **Nitrate.** Spring S-7 at Mandi had 37 and 36 mg/L of nitrate, and bore well B-3 at Rewalsar had 32 and 27.9 mg/L, in May and November. These are below the limit of 45 mg/L, but well above most other sites. Both are in crowded towns, and the likely sources are sewage and farm runoff.

For irrigation the water is safe. SAR is below about 1.1 in every sample, average percent sodium is about 11–16%, and RSC is below 1.25 meq/L, so the risk of sodium or salt damage to soil is low (Richards, 1954). Six samples in each season had a corrosivity ratio above 1, which can shorten the life of metal pipes.

6. Recharge Planning and Management

6.1 Choosing sites for recharge structures

Sharma et al. (2025) chose recharge sites in three steps:

- Points where lineaments cross streams were taken as first candidates.
- Points on steep slopes were removed. Only the low and moderate slope classes of their slope map were kept; on our reading of the class limits (0–9°, 9–18°, 18–27°, 27–40° and above 40°), this means slopes of roughly 18° or less.

- The remaining points were checked against land use and groundwater potential, and sites near crowded foothill settlements were given priority.

Steep sites were avoided for two reasons: water runs off too fast to be held, and the weight of a structure on a steep slope can lead to slope failure. They add that field surveys are still needed to fix each site and judge its cost.

6.2 Choosing the type of structure

Pophare and Balpande (2015) matched structures to stream order across the 53 small watersheds of the Suketi basin, following national guidelines (CGWB, 2007):

- **First- and second-order streams:** gully plugs and gabion structures.
- **Second- and third-order streams:** nala bunds and percolation tanks.
- **Fourth- to seventh-order streams:** check dams and subsurface dykes, where land and local conditions allow.
- **Spring catchments:** about three check dams or gabions per spring, or about 282 structures in all, each about 15 m wide and 1.5 m high.

They also suggested rooftop rainwater harvesting, farm ponds and small underground tanks in the hills. In the valley, they advised recharge where the post-monsoon water level is deeper than about 3 m, and cleaning and deepening of old village ponds. For the Gaj watershed, Sharma et al. (2025) recommended check dams, gabions, contour trenches, pony-pit wells and roof-water tanks.

6.3 Traditional water systems

Pophare and Balpande (2015) described several traditional systems still found in the Suketi basin: *baudis* (stone-lined pits around a spring), *nauns* (larger covered spring tanks), *chhrudus* (piped spring outlets) and *kuhls* (channels that carry stream water to fields) (Sharma and Kanwar, 2009). Many are now neglected because of piped supply. In the Gaj watershed, Sharma et al. (2025) mention similar spring structures, known locally as *bowris*, which can go dry outside the rainy season. A recent field study in Kangra selected two such springs for study and revival in 2021–2022 and found their water within national quality limits (Rawal et al., 2024). Repairing them is a low-cost addition to new recharge works.

7. Critical Assessment and Research Gaps

The three studies are most useful when read together, because each covers a different part of the problem. Their main limitations are as follows.

- **Short records.** Both Suketi studies rely on one year of data (2009), and the resource figures date from 2004. They show the seasonal pattern but not long-term change.
- **No field check of the Gaj maps.** The study had no observation-well or spring-discharge data. Its maps were compared with an infiltration map, but that map was built from five layers (soil, slope, temperature, land use and curvature) that also enter the potential model. Close agreement between the two is therefore partly expected. Testing against measured spring flow or well levels would be stronger (Machiwal et al., 2011).
- **Quality and recharge kept apart.** The Gaj study has no chemical data, although its catchment also contains granite and growing towns. Recharge structures in crowded foothills could carry polluted runoff into the aquifer, so water quality should be tested before sites are fixed.
- **No water budget.** Neither study estimates how much water the proposed structures would add, what they would cost, or who would maintain them. These factors often decide whether recharge schemes succeed (Bouwer, 2002).

Future work could address these gaps in the following ways:

1. **Monitoring.** Set up observation wells and gauged springs in both catchments and measure them every season.

2. **Isotopes.** Use stable isotopes of oxygen and hydrogen to find the height and location of spring recharge areas and to separate rain from snowmelt (NITI Aayog, 2018).
3. **Geophysics.** Use electrical resistivity surveys to check the thickness of soil and weathered rock before building structures.
4. **Modelling.** Build groundwater flow models for the Balh valley to test pumping and recharge options under different rainfall conditions.
5. **Wider quality data.** Sample the Gaj watershed, and add trace metals and bacteria to future surveys in both areas.
6. **People and costs.** Study land ownership, community willingness, upkeep and costs and benefits of each proposed structure, and plan the revival of traditional systems with local users (Tambe et al., 2012).

8. Conclusions

Groundwater in these two Himalayan catchments occurs in two very different settings. Valley fill such as the Balh valley holds a large and little-used resource with shallow water levels, while the hills depend on small springs that weaken before every monsoon. The groundwater of the Suketi basin is generally good for drinking and irrigation, but fluoride in granite areas and nitrate in towns call for regular testing. In the Gaj watershed, the zones most favourable for groundwater became smaller between 2010 and 2022 as rainfall declined, and the best recharge sites are on gentle to moderate slopes near settlements.

Each study covers one part of the problem: quantity, quality or site selection. Future work should combine them, check maps against field data, test water quality before building structures, and involve local communities, including in the care of traditional water systems. This would give planners in the Himalayan foothills a sounder basis for long-term water security.

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Conflict of interest: The authors declare no competing interests.

Data availability: All data discussed are taken from the published sources cited.

References

- APHA (1998). *Standard Methods for the Examination of Water and Wastewater*, 20th edn. American Public Health Association, Washington, D.C.
- BIS (2012). *Indian Standard Drinking Water – Specification (Second Revision), IS 10500:2012*. Bureau of Indian Standards, New Delhi.
- Bouwer, H. (2002). Artificial recharge of groundwater: hydrogeology and engineering. *Hydrogeology Journal*, 10(1), 121–142.
- CGWB (2006). *Ground Water Information Booklet, Mandi District, Himachal Pradesh*. Central Ground Water Board, North Western Himalayan Region, Dharamshala.
- CGWB (2007). *Manual on Artificial Recharge of Ground Water*. Central Ground Water Board, Ministry of Water Resources, Government of India, New Delhi.
- Gibbs, R.J. (1970). Mechanisms controlling world water chemistry. *Science*, 170(3962), 1088–1090.
- Kumar, S. and Mahajan, A.K. (2001). Seismotectonics of the Kangra region, northwest Himalaya. *Tectonophysics*, 331(4), 359–371.
- Machiwal, D., Jha, M.K. and Mal, B.C. (2011). Assessment of groundwater potential in a semi-arid region of India using remote sensing, GIS and MCDM techniques. *Water Resources Management*, 25(5), 1359–1386.
- Meinzer, O.E. (1923). *Outline of Ground-water Hydrology, with Definitions*. U.S. Geological Survey Water-Supply Paper 494, Washington, D.C.

- NITI Aayog (2018). *Inventory and Revival of Springs in the Himalayas for Water Security*. Report of Working Group I, Government of India, New Delhi.
- Piper, A.M. (1944). A graphic procedure in the geochemical interpretation of water-analyses. *Transactions, American Geophysical Union*, 25(6), 914–928.
- Pophare, A.M. and Balpande, U.S. (2014). Morphometric analysis of Suketi river basin, Himachal Himalaya, India. *Journal of Earth System Science*, 123(7), 1501–1515.
- Pophare, A.M. and Balpande, U.S. (2015). Groundwater resource development and management in Suketi river basin, Himachal Himalaya, India. In: Shrivastava, K.L. and Srivastava, P.K. (Eds.), *Frontiers of Earth Science*. Scientific Publishers (India), 499–526.
- Pophare, A.M., Balpande, U.S. and Nawale, V.P. (2018). Hydrochemistry of groundwater in Suketi river basin, Himachal Himalaya, India. *Journal of Geosciences Research*, 3(1), 67–83.
- Rawal, S., Jose, P.G., Singh, D. and Linda, A. (2024). Ensuring water security by reviving selected springs in Kangra region, Himachal Himalaya, India. *Water Science*, 38(1). <https://doi.org/10.1080/23570008.2024.2333600>
- Richards, L.A. (Ed.) (1954). *Diagnosis and Improvement of Saline and Alkali Soils*. Agriculture Handbook No. 60, United States Department of Agriculture, Washington, D.C.
- Rodell, M., Velicogna, I. and Famiglietti, J.S. (2009). Satellite-based estimates of groundwater depletion in India. *Nature*, 460(7258), 999–1002.
- Saaty, T.L. (1980). *The Analytic Hierarchy Process*. McGraw-Hill, New York.
- Sharma, N. and Kanwar, P. (2009). Indigenous water conservation systems – a rich tradition of rural Himachal Pradesh. *Indian Journal of Traditional Knowledge*, 8(4), 510–513.
- Sharma, S., Tamrakar, Y., Deep, A. and Mahajan, A.K. (2025). Multi-criteria assessment and decision-making approach for planning of groundwater recharge structures in a highland watershed. *Discover Sustainability*, 6, 1163. <https://doi.org/10.1007/s43621-025-02054-3>
- Somers, L.D. and McKenzie, J.M. (2020). A review of groundwater in high mountain environments. *WIREs Water*, 7(6), e1475.
- Srikantia, S.V. and Bhargava, O.N. (1998). *Geology of Himachal Pradesh*. Geological Society of India, Bangalore.
- Tambe, S., Kharel, G., Arrawatia, M.L., Kulkarni, H., Mahamuni, K. and Ganeriwala, A.K. (2012). Reviving dying springs: climate change adaptation experiments from the Sikkim Himalaya. *Mountain Research and Development*, 32(1), 62–72.
- Valdiya, K.S. and Bartarya, S.K. (1989). Diminishing discharges of mountain springs in a part of Kumaun Himalaya. *Current Science*, 58(8), 417–426.
- WHO (2017). *Guidelines for Drinking-water Quality: Fourth Edition Incorporating the First Addendum*. World Health Organization, Geneva.